WHO'S WHO · PAGE 1 OF 2

Every name I've used so far, and what it is

Colored tag = which kind of thing it is. You don't need to memorize these. They're the "for example" behind each category, so when a name shows up you know what bucket it belongs in.

LIVE **Laika** — a Moscow stray dog, the first animal to orbit Earth, inside Sputnik 2 (1957). Proved a living body survives launch and weightlessness. Didn't survive the flight.

LIVE **ISS** — International Space Station. A football-field-sized lab in LEO (~400 km) where 6–7 astronauts live for months. Built by the US, Russia, Europe, Japan, Canada. Orbits every 90 minutes.

LOOK OUT **Hubble** — a bus-sized telescope orbiting Earth since 1990 at ~540 km. Above the air, so razor-sharp pictures. Source of most famous deep-space photos.

LOOK OUT **Kepler** — a space telescope (2009–2018) that stared at one patch of sky and found thousands of planets around other stars.

LOOK DOWN **GOES** — US weather satellites in GEO. The cloud loops on the TV weather report are GOES pictures.

RELAY **Starlink** — SpaceX's internet constellation: thousands of small satellites in LEO (~550 km) relaying internet to dishes on the ground.

FLYBY **Voyager 1 & 2** — twin NASA probes launched 1977. Flew past Jupiter and Saturn (Voyager 2 also Uranus and Neptune), then kept going. Both have now left the solar system and are still transmitting.

LIVE **Sputnik 2** — the Soviet satellite that carried Laika. Sputnik 1, a month earlier, was the very first artificial satellite of any kind: a beeping metal ball.

LIVE **Tiangong** — China's own space station, smaller than ISS, also in LEO.

LOOK OUT **JWST** — James Webb Space Telescope (2021). Hubble's bigger successor, sees infrared (heat-light). Parked 1.5 million km from Earth at a gravity balance point, not in Earth orbit.

LOOK DOWN **Landsat** — US series of Earth-mapping satellites running since 1972. Farms, forests, cities, floods, all photographed from LEO.

RELAY **GPS** — Global Positioning System: ~30 US satellites in MEO (~20,200 km) broadcasting time signals. Your phone hears several and computes where you are. Europe's version is Galileo.

RELAY **Queqiao** — a Chinese relay satellite parked beyond the Moon so a lander on the Moon's far side (Chang'e-4) could talk to Earth, since the Moon blocks direct radio.

FLYBY **New Horizons** — NASA probe launched 2006, flew past Pluto July 2015 after 9.5 years, took the famous heart-shaped-glacier photos, then continued into the Kuiper Belt.

Orbiters, probes, landers, rovers, and the places

Second half. Places at the bottom, because a few of them (Titan, Pluto, Kuiper Belt) came up as destinations.

ORBITER **Cassini** — NASA spacecraft that orbited Saturn 2004–2017, photographing the rings and moons. Deliberately crashed into Saturn at the end so it couldn't contaminate a moon.

ATMOSPHERIC **Galileo probe** — dropped by the Galileo orbiter into Jupiter's atmosphere in 1995. Transmitted for 58 minutes before heat and pressure crushed it. No surface to land on.

ORBITER **MRO** — Mars Reconnaissance Orbiter (2006–). Maps Mars in high detail AND relays radio for the rovers on the ground. Orbiter and relay at once.

LANDER **InSight** — NASA Mars lander (2018–2022). Planted a seismometer and listened for marsquakes. Died when dust covered its solar panels.

ROVER **Sojourner** — the first Mars rover (1997), microwave-oven sized.

ROVER **Curiosity & Perseverance** — car-sized, nuclear-powered Mars rovers (2012 and 2021). Perseverance carried the Ingenuity helicopter and is collecting samples for a future return.

PENETRATOR **LCROSS** — NASA mission (2009) that crashed a spent rocket stage into a dark Moon crater and flew through the dust plume. Found water ice.

Pluto — dwarf planet at the edge of the solar system, ~33 times farther from the Sun than Earth. Visited once, by New Horizons.

L2 — a gravity balance point 1.5 million km behind Earth (away from the Sun) where a spacecraft can hover with little fuel. JWST lives there.

Walter Hohmann / Johann Lambert — not missions, people. Hohmann: German engineer who worked out the cheapest two-burn transfer in 1925. Lambert: 18th-century mathematician whose "two points plus a time" orbit problem still carries his name.

GLONASS / BeiDou — Russia's and China's navigation constellations, both in MEO like GPS. Your phone usually listens to all of them.

ATMOSPHERIC **Huygens** — a European probe Cassini carried and dropped onto Titan (Saturn's biggest moon) in 2005. Parachuted through the orange haze, landed, sent pictures for 90 minutes. One mission, two types.

ORBITER **Juno** — NASA Jupiter orbiter, arrived 2016. The huge solar arrays in your clean-room slide are Juno's; it's the farthest solar-powered spacecraft ever.

LANDER **Viking 1 & 2** — NASA's first successful Mars landers (1976). Sat in one spot, photographed the surface, tested the soil for life.

LANDER **Rosetta / Philae** — Rosetta = European spacecraft that orbited a comet (2014). Philae = the small lander it dropped onto the comet's surface. Philae bounced twice, landed in a shadow, ran out of solar power in 3 days.

ROVER **Spirit & Opportunity** — twin golf-cart-sized Mars rovers (2004). Designed for 90 days; Opportunity drove for 14 years. The rover in your slide is this design.

PENETRATOR **Deep Impact** — NASA mission (2005) that fired a washing-machine-sized copper block into comet Tempel 1 and watched what sprayed out.

Titan — Saturn's largest moon. Thick orange atmosphere, lakes of liquid methane. The only moon anyone has landed a probe on (Huygens).

Kuiper Belt — the ring of icy leftovers beyond Neptune. Pluto is its biggest known member.

Comet 67P / Churyumov-Gerasimenko — the duck-shaped comet Rosetta orbited from 2014 and Philae landed on. Named after the two astronomers who spotted it in 1969.

Galileo (two things) — the European navigation constellation (GPS rival, MEO), AND NASA's 1990s Jupiter orbiter that dropped the atmospheric probe. Context tells you which.

NASA / ESA / SpaceX — US space agency / European Space Agency / the private US rocket company (Falcon rockets, Starlink). Not affiliated with your notebook title.